Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Project Update

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Agenda

Background. The PDAC prognostic problem, and why to supervise the factorization

Method. The joint model, explicit priors, the ELBO, CAVI inference, preprocessing, and a new two-stage approach to selecting K

Results. A synthetic validation of the K-selection approach, then real PDAC data: gene programs, survival, and the joint model vs. the unsupervised two-step

Impact & Future Directions

Roadmap for the next ~15-20 minutes. K selection works in two stages: a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index picks K_init, and ARD shrinkage then determines which of those factors are actually survival-active (K_eff). The projection predictor (YF) is the only model presented.

Background

Background: PDAC needs molecular prognosis

- **Five-year survival under 12%**, among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals, so prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs (GEPs)* whose activation predicts survival, and validate them on **independent** cohorts
- A GEP is a coordinated set of genes (a column of F) whose patient-level activation (a column of L) we test for survival association

ℹ Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance**, often batch or housekeeping signal
- A program explaining little variance but strongly prognostic gets **missed**

- **Supervision** steers factors toward survival-relevant structure *during* estimation

A program explaining 15% of variance may be pure batch effect; one explaining 3% may cleanly separate survivors. Supervision prioritizes the second.

One background slide. The clinical motivation is unchanged. The methodological motivation, joint over two-step, is the thread of the whole talk, and I back it with both a synthetic comparison and a real-data external comparison against the unsupervised two-step (EBMF → Cox) baseline.

Methods

The model: two coupled components

$$\underbrace{Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \qquad \underbrace{h_i(t) = h_0(t)\exp(\eta_i)}_{\text{Cox proportional hazards}}$$

Quantity	Dim	Meaning
Y	$n \times p$	expression matrix
L	$n \times K$	patient program-activation scores
F	$p \times K$	gene weights per program
β	$K \times 1$	survival (log-HR) coefficients
τ_j	scalar	gene-specific noise precision
η_i	scalar	patient log-risk score, $(Y_i F)\beta$

Shared F couples the two

The **same** gene-program weights F drive both the low-rank reconstruction of Y **and** the survival risk score $\eta = (YF)\beta$. A program is **prognostic only if its** $\beta_k \neq 0$; otherwise it explains expression alone.

Non-parametric baseline hazard

$h_0(t)$ is left **unspecified**; it cancels in the Cox partial likelihood. No Weibull / exponential shape assumption, only proportional hazards.

Two equations, one shared latent structure. Every symbol is defined in the table, including the risk score eta, which is now always the projection score (YF) . The baseline hazard is non-parametric, so h0(t) drops out of the partial likelihood and we make no parametric shape assumption.

The model: projection predictor and explicit priors

Projection predictor: $\eta = (YF)\beta$

- Risk uses the **direct genomic projection** YF, not a re-derived patient score

- **Exact** out-of-sample scoring: $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$ is the *identical* formula at train and test, so there is no train/test score mismatch
- L is fit for the genomics reconstruction only; it does not enter the survival likelihood

> Why the projection form
>
> Leakage-free **exact** external scoring, with no OLS re-projection step for new patients. This is what makes external validation on held-out cohorts a clean apples-to-apples comparison.

Priors: every block is explicit. $q(\cdot)$ denotes the **variational (approximate posterior)** for each block, fit by empirical Bayes at every update:

- $q(L_{\cdot k})$, $q(F_{\cdot k})$: **point-exponential** (non-negative spike-and-slab), $g(\theta) = \pi_0\delta_0 + (1-\pi_0)\,\mathrm{Exp}(\lambda)$, $\theta \geq 0$
- $q(\beta_k)$: **normal** (Gaussian slab, no spike at zero), $g(\beta) = N(0, \sigma^2)$
- $q(\tau_j)$: **closed-form MLE**, no EBNM, no prior tuning

Sparsity π_0, scale λ, and slab variance σ^2 are all estimated from the data (`ebnm`) at each step.

> Why a normal prior on β, not a spike-and-slab
>
> A spike-and-slab prior forces a hard in/out decision on each factor as part of fitting. A normal prior instead lets $\hat{\beta}_k$ shrink **continuously** toward zero for a factor with no survival signal, without ever forcing it to exactly zero. That continuous shrinkage is what Automatic Relevance Determination (ARD) means here, and it is the raw material Stage 2 of K selection (two slides ahead) turns into a discrete active/inactive call, by thresholding $|\hat{\beta}_k|$ after fitting rather than building the threshold into the prior itself.

The projection predictor (YF) scores new patients with the identical formula used at train time, so there is no mismatch and no re-derivation. Priors are explicit: non-negative point-exponential on L and F, a Gaussian (normal) prior on , and a closed-form MLE for . The normal-not-spike-and-slab choice on   is worth landing clearly here: it gives continuous shrinkage during fitting, and the hard active/inactive call happens afterward, by thresholding, which is exactly the ARD mechanism the K-selection slide relies on next.

The objective: ELBO & variational family

Mean-field variational family: each block factorizes independently:

$$q(L, F, \beta, \tau) = \prod_{k=1}^{K} q(L_{\cdot k}) \prod_{k=1}^{K} q(F_{\cdot k}) \prod_{k=1}^{K} q(\beta_k) \prod_{j=1}^{p} q(\tau_j)$$

The **ELBO** (evidence lower bound) is the objective CAVI maximizes, a lower bound on the log marginal likelihood. For the projection model:

$$\mathrm{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda\, E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\textstyle\sum_{\text{blocks}} D_{\text{KL}}(q \,\|\, \text{prior})}_{\text{prior regularization}}$$

- **Genomics fit** $E_q[\mathcal{L}_{\text{gen}}] = \sum_j E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$: Gaussian reconstruction of Y by $LF^\top$

- **Survival fit**, the projection predictor written out explicitly:

$$E_q[\mathcal{L}_{\text{Cox}}] \approx \log PL(\eta_0) - \tfrac{1}{2}\sum_i w_i \sum_k (YF)^2_{ik}\,\text{Var}_q(\beta_k), \quad \eta_0 = (YF)\,E_q[\beta]$$

  a **2nd-order Taylor** expansion of the Cox partial log-likelihood around η_0, the point estimate of $(YF)\beta$. Because YF is treated as data (not a random quantity with its own posterior variance) once F is fixed, only β's posterior uncertainty survives in the variance term. This is what keeps the survival term conjugate to the EBNM updates.

- **KL term** penalizes each posterior q away from its prior: the adaptive shrinkage
- The **shared** F enters both likelihood terms, genomics reconstruction ($Y \approx LF^\top$) and survival risk ($\eta = (YF)\beta$); that coupling *is* the supervision
- The ELBO **balances** expression and survival automatically, with **no tuning parameter** α ($\lambda = 1$ in the recommended fits)

  What the ELBO is *for*: it is the **fitting objective**. CAVI maximizes it, and a fit is declared converged when it stops improving. It is also **one of four criteria**, alongside BIC, log-likelihood, and cross-validated external C-index, used to choose K_{init} (next: *Selecting* K).

The variational family is mean-field and factor-wise. The ELBO has three pieces: Gaussian genomics reconstruction, the Cox survival term via a second-order Taylor expansion that keeps the updates conjugate, and a per-block KL penalty that is the adaptive shrinkage. I'm showing the survival term spelled all the way out this time, with (YF)beta substituted for eta, so it's explicit rather than abstract: eta_0 is the point estimate (YF) times E_q[beta], and the variance term only involves beta's own posterior uncertainty, because YF is fixed data given the current F, not something with its own posterior spread. The shared F in both likelihood terms is the supervision. The ELBO is one of the four K_init-selection criteria alongside BIC, log-likelihood, and cross-validated external C-index.

Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel): per outer iteration, for each factor k:

$$\beta_k \;\to\; L_{\cdot k} \;\to\; F_{\cdot k}, \qquad \text{then } \tau \text{ once per sweep}$$

- Each update is an **EBNM** sub-problem: pseudo-data $x = B/A$, scale $s = 1/\sqrt{A}$, then `ebnm` returns the posterior mean **and** variance
- β_k has precision governed by the projection score's own variance across patients, so a factor with **no survival signal** shrinks $\beta_k \to 0$ on its own: the mechanism ARD relies on
- The empirical-Bayes prior $\hat{g}$ is **re-estimated every update**, giving adaptive shrinkage per factor and per gene; the residual is deflated after each factor

Convergence. After a 5-iteration burn-in, stop when the **ELBO stops improving**: its relative change between iterations falls below 10^{-5}:

$$\frac{\left|\text{ELBO}^{(t)} - \text{ELBO}^{(t-1)}\right|}{\left|\text{ELBO}^{(t-1)}\right|} < 10^{-5}$$

- The ELBO is the **single objective** being maximized, so this directly detects a stationary point of the variational optimization
- Mean absolute changes in L and β are tracked alongside as **diagnostics** (both settle below 10^{-3} at convergence)

! Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at $C < 0.5$. **Fix:** after convergence, orient β against the **training** concordance (flip its sign if $C_{\text{train}} < 0.5$), and carry that same orientation into every downstream use of β, including test scoring.

CAVI updates one factor at a time in Gauss-Seidel order; each block reduces to an EBNM call with pseudo-data x=B/A and scale s=1/sqrt(A). Convergence requires both L and beta to settle below 1e-3 in MEAN absolute change; averaging avoids a single oscillating factor blocking the stop. The sign correction orients beta by training concordance once, after convergence, and that same orientation carries through to every later use, including test scoring – it is not reapplied at test time separately. If asked whether this is fully resolved: it's the same design as before, not replaced, and it has one known minor caveat noted in DECISIONS.md – the orientation check pools all patients rather than respecting stratification when strata are used elsewhere in the model. The effect is documented as negligible. (Separately, the project's old 5-fold cross-validation K-selection function did disable this per fold, since a per-fold flip would have made C-index comparable across folds meaningless – but that function isn't part of the current K_init sweep, which fits once per K_init on the full training set and scores on external cohorts, not folds.)

Preprocessing is crucial: per-platform normalization

The $p \gg n$ scaling problem. Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. The survival precision for program k is

$$A_k = \sum_i w_i \, (YF)_{ik}^2,$$

where w_i = the patient's **Cox working weight** (curvature from the survival Taylor step) and $(YF)_{ik}$ = patient i's **projection score** on program k. Without correction, between-platform variance inflates (YF) unevenly, **swamps** A_k, and the survival coefficients **collapse to** $\beta \to 0$.

The fix: normalize *per platform, before merging*:

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean × variance), top-3000 per cohort, then **intersect** → **2064 genes** in the merged training matrix
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

! Skip per-platform normalization → collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive $\beta \to 0$.

No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

The single most important practical finding since April. A_k is the survival precision for program k. Without per-platform normalization the projection scales blow up across platforms, A_k is dominated by between-platform variance, and beta collapses. The fix is per-platform z-standardization before merging, which leaves 2064 genes. 10 of 12 alternatives collapse. Normalization is fit per training cohort before any split, so there is no leakage.

Selecting the number of programs: a two-stage framework

Stage 1: K_{init}, by a consensus of four fit/prediction criteria. Fit at several K_{init}; compare ELBO, BIC, an ELBO-style joint log-likelihood, and cross-validated external C-index across the grid.

Stage 2: the active-factor subset, by ARD. Fit once at the chosen K_{init}; the normal prior on β (previous slide) lets Automatic Relevance Determination shrink unneeded factors' $\beta_k \to 0$ **on its own**, without a second per-K validation loop.

Stage 1: choosing K_{init}

- **ELBO**: the fitting objective itself, compared *across* K_{init}
- **BIC** $= -2\,\text{LL} + \log(n) \cdot \text{df}$, $\text{df} = K_{\text{init}} \cdot (n + p + 1)$, a complexity penalty that is **not** guaranteed to agree with ELBO or external C
- **Log-likelihood**: genomics + survival, an ELBO-style bound (not an exact marginal likelihood)
- **Cross-validated external C-index**: the only *generalization* signal among the four, evaluated on the five held-out cohorts
- These need **not** agree; see *Results* for the full $K_{\text{init}} = 2\text{-}20$ sweep, with each metric's own direction of improvement made explicit

Stage 2: ARD classification at fixed K_{init}

Each factor k is classified by `classify_factors()`:

- **survival-active**: $|\hat{\beta}_k| > 0.001$
- **genomics-only**: not survival-active, but $\text{PVE}_k > 1\%$
- **dead**: neither, fully pruned

Running example: $K_{\text{init}} = 7$

7 starting factors → **2 survival-active** + **2 genomics-only** (4 kept, 3 pruned). Stable across a wide range of K_{init}; see *Results*.

Two stages. Stage 1 picks K_init by a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index – these don't have to agree, and the Results section shows the full sweep, with each metric's preferred direction stated explicitly, and where they disagree. Stage 2 is ARD: at a fixed K_init, the normal prior on beta lets survival-irrelevant factors shrink to beta=0 on their own, and classify_factors() splits the rest into survival-active (|beta|>0.001) vs genomics-only (PVE>1%) vs dead. K_init=7 is the running example throughout: 2 survival-active, 2 genomics-only, 4 kept total, 3 pruned, the same real-data configuration used throughout Results.

Results

Data: Synthetic Validation & PDAC Cohorts

Multi-cohort simulation

EBMF-grounded ground truth: shared factors (present in all studies) plus study-specific factors. **Shared factors carry the survival signal**; study-specific factors do not.

Scenarios

- *All Shared Factors*: every program is shared and prognostic
- *All Cohort-Specific Factors*: negative control, no shared signal ($\eta \equiv 0$)
- *Shared & Cohort-Specific Factors*: the realistic mix

Model Arms

- Unsupervised two-step (EBMF → Cox)

- The projection-predictor joint model, K selected by over-specified K_{init} plus ARD pruning, mirroring the real-data procedure

PDAC cohorts: External Validation

- **Train:** TCGA-PAAD + CPTAC ($n \approx 273$, RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the **exact** projection formula $\eta = (Y_{\text{new}} F)\beta$

Held-out cohort	Platform
Dijk	RNA-seq
Moffitt	Microarray
PACA-AU (array)	Microarray
PACA-AU (seq)	RNA-seq
Puleo	Microarray

Two datasets, simulation first, then real PDAC data, in that order for the rest of Results. The simulation now selects K the same way the real-data procedure does: an over-specified K_init pruned by ARD, instead of the oracle true-K used in June, and compares only the joint model against the unsupervised EBMF-then-Cox baseline. On real data we train on TCGA + CPTAC and validate on five fully held-out cohorts spanning microarray and RNA-seq, scored by the exact projection formula.

Synthetic Results: over-specifying K_{init} doesn't hurt performance

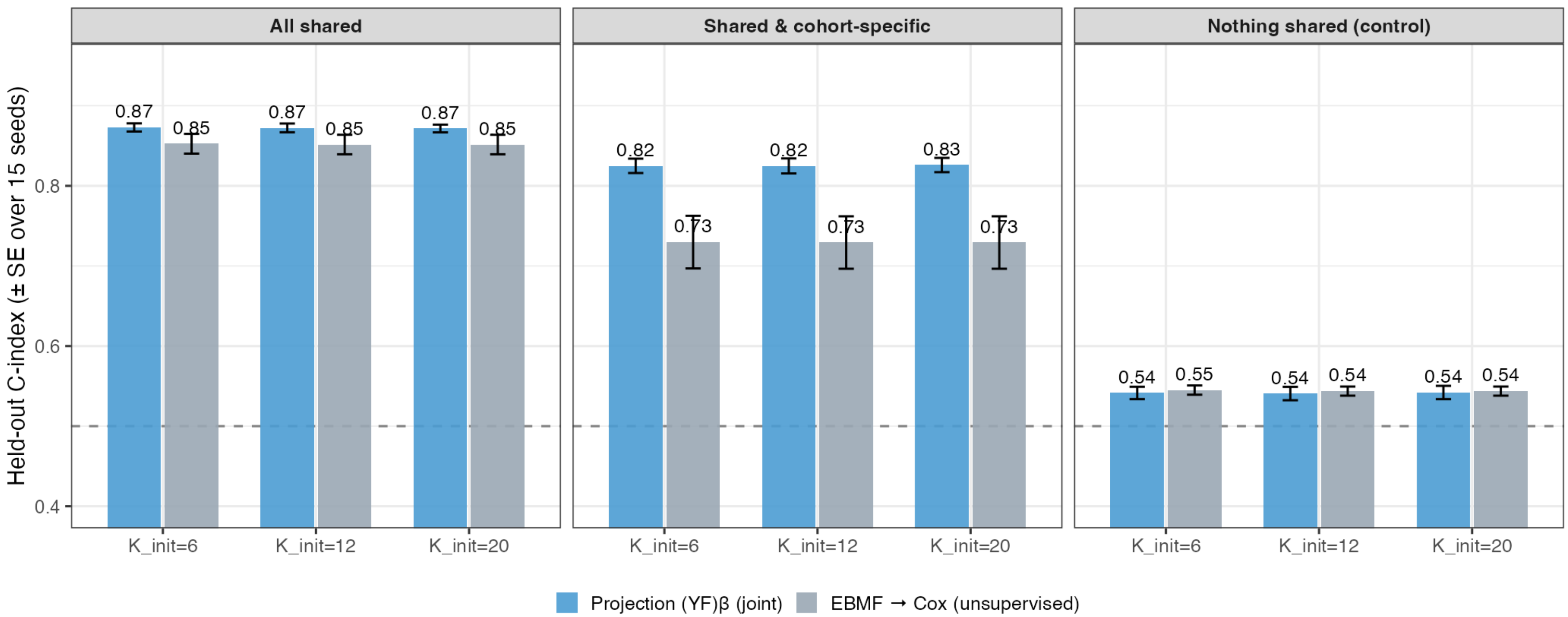

Figure 1: **Figure.** Held-out C-index by K_{init}, arm, and scenario (mean ± SE over 15 seeds; dashed line = chance).

- $K_{\textbf{init}} = 6$ means the model was told the true number of factors directly (an idealized case used only as a reference point, never available on real data); $K_{\textbf{init}} = 12/20$ is the over-specified-plus-ARD-pruning procedure actually used
- **C-index (and, in the appendix-detail figure, factor recovery and specificity) are flat across** $K_{\textbf{init}} = 6 \to 12 \to 20$ in every scenario: giving the model more starting factors than it needs and trusting ARD to prune costs nothing relative to knowing the true count in advance
- Bars are the mean over 15 seeds (up from 5, for a properly powered comparison); error bars are $\pm$ 1 standard error across those 15 seeds

The simulation now uses the real-data K-selection procedure: over-specify K_init, let ARD prune, instead of June's oracle true-K. K_init=6 is kept only as a reference point, since on real data we never actually know the true K in advance. The headline is a clean null result: C-index doesn't move at all across K_init=6/12/20, in any of the three scenarios, with 15 seeds behind each bar. Over-specifying K and trusting ARD costs nothing. The paired recovery and specificity figures, which tell the same flat story, are in the appendix.

Synthetic Results: does a larger K_{init} create more false alarms?

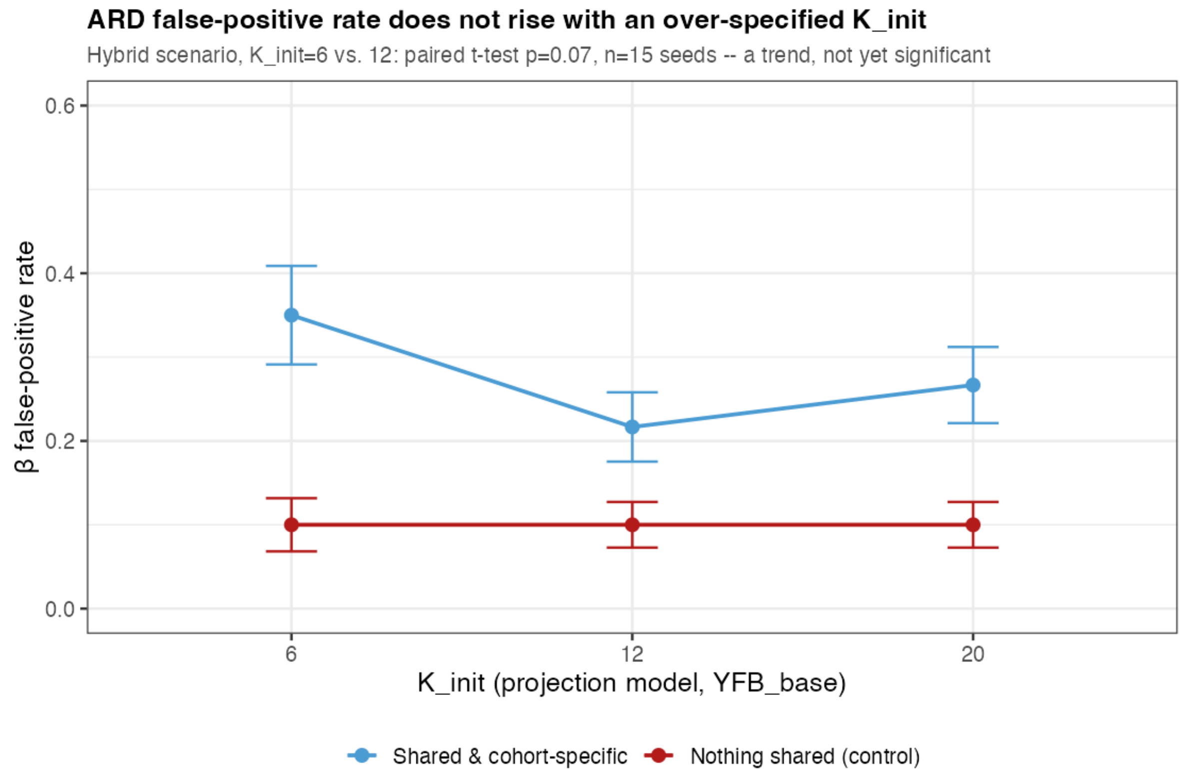

Figure 2: **Figure.** β false-positive rate: how often a factor with *no* true survival effect is nonetheless called survival-active, vs. K_{init} (projection model only; EBMF has no survival coefficient to test this on). Mean $\pm$ SE over 15 seeds.

- **The question:** giving the model more factors than it needs gives ARD more non-signal factors it *could* mistakenly call survival-active. Does the false-positive rate actually rise as K_{init} grows?
- **The answer, so far: no.** In the realistic scenario the false-positive rate is numerically *lower* at $K_{\text{init}} = 12/20$ than at $K_{\text{init}} = 6$ ($0.35 \rightarrow 0.22 \rightarrow 0.27$), though a paired test between $K_{\text{init}} = 6$ and 12 gives $p = 0.07$, so call this a consistent trend, not yet a statistically confirmed one. In the negative-control scenario the rate is flat at 0.10 regardless of K_{init}.
- Consistent with a separate point raised in discussion: even when a non-signal factor does pick up a small, nonzero coefficient, that coefficient is too small to meaningfully move the risk score

This is the one place a genuinely new question could have shown up: does over-specifying K produce MORE false-positive survival-active calls, since ARD has more non-signal factors to potentially mis-activate. At 5 seeds it looked like a clear improvement; at 15 seeds the direction holds in the realistic scenario but the paired test is p=0.07, not significant, and the negative control is completely flat. I'm presenting this as what it is, a promising trend, not a confirmed finding, and tying it back to the point raised after the last simulation exercise: a small spurious coefficient on a non-signal factor doesn't meaningfully move the risk score, which is consistent with false positives here not being large in magnitude even when they occur.

Joint model vs. two-step as survival signal strength varies

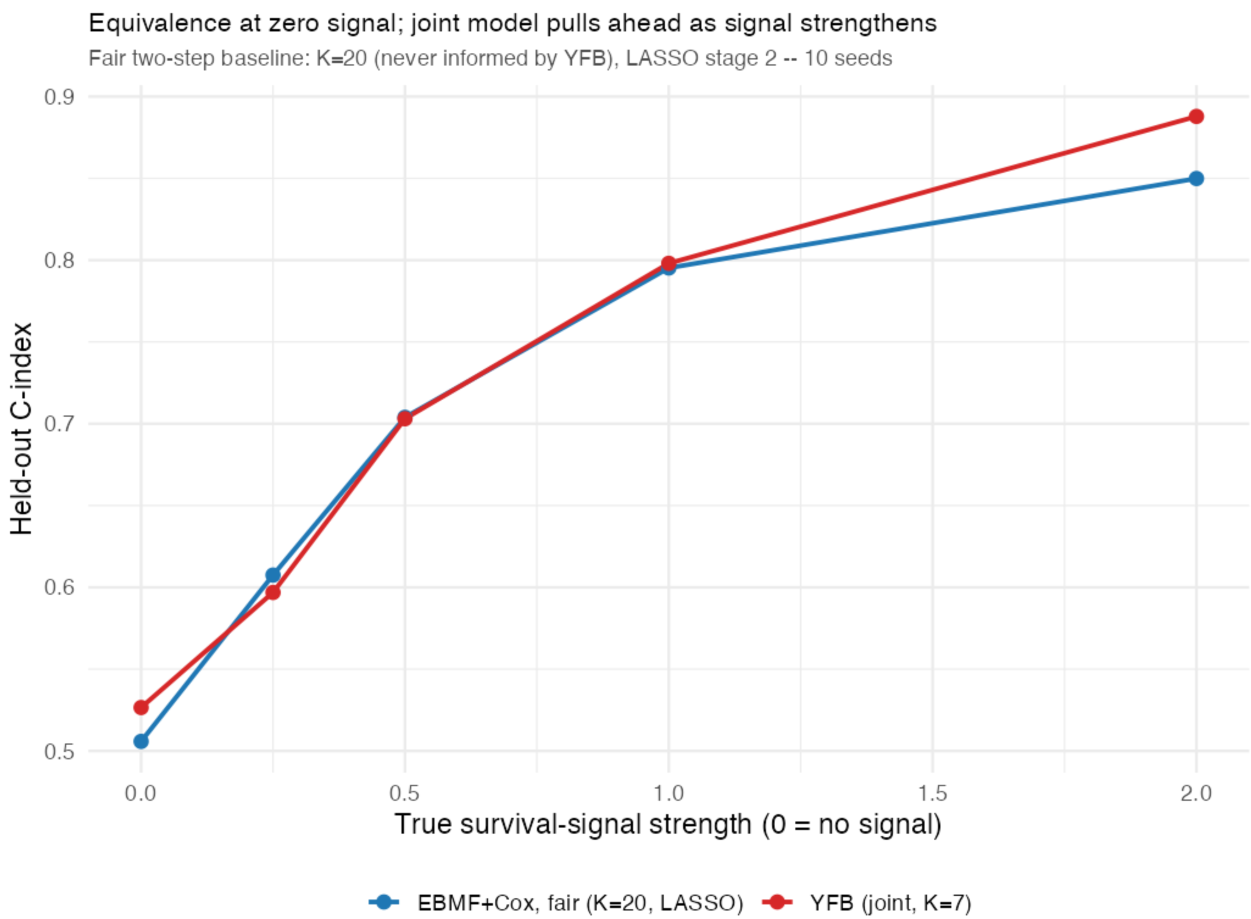

Figure 3: **Figure.** External performance of the joint model vs. the unsupervised two-step (EBMF → Cox) as the simulated survival-signal strength increases. "Strength" is a multiplier on the true survival coefficients used to generate each simulated patient's survival time (values swept: 0, 0.25, 0.5, 1, 2; strength = 1 is this project's standard "realistic-magnitude" setting). At strength = 0 (no true survival signal at all), the joint model matches EBMF: supervision costs nothing when there is nothing to exploit. At strength = 2 (twice the realistic magnitude), the joint model pulls ahead.

- **Strength = 0:** joint ≈ EBMF-only. The joint model does not fabricate prognostic signal when none exists.
- **Strength = 2:** joint model outperforms the two-step. Supervision helps *only when there is signal to find*, and does not hurt when there isn't.
- With only 10 replicate seeds, no single strength level individually reaches statistical significance; the pattern across the sweep is the evidence, not any one point

This single figure carries two points. Strength is a literal multiplier on the true survival coefficients used to simulate outcomes, swept at 0, 0.25, 0.5, 1, and 2, with 1 being the project's standard realistic magnitude, so strength 2 means twice that. At zero simulated survival signal, the joint model tracks EBMF almost exactly: no false discovery, no manufactured prognostic signal. As signal strength increases, the joint model pulls ahead of the two-step baseline. Together this is the cleanest argument that supervision is a free option: it costs nothing when uninformative and helps when it isn't. Worth noting if asked: only 10 seeds, so no individual strength value is individually significant; it's the trend across the whole sweep that carries the argument.

PDAC Cohorts: Prognostic Gene Programs

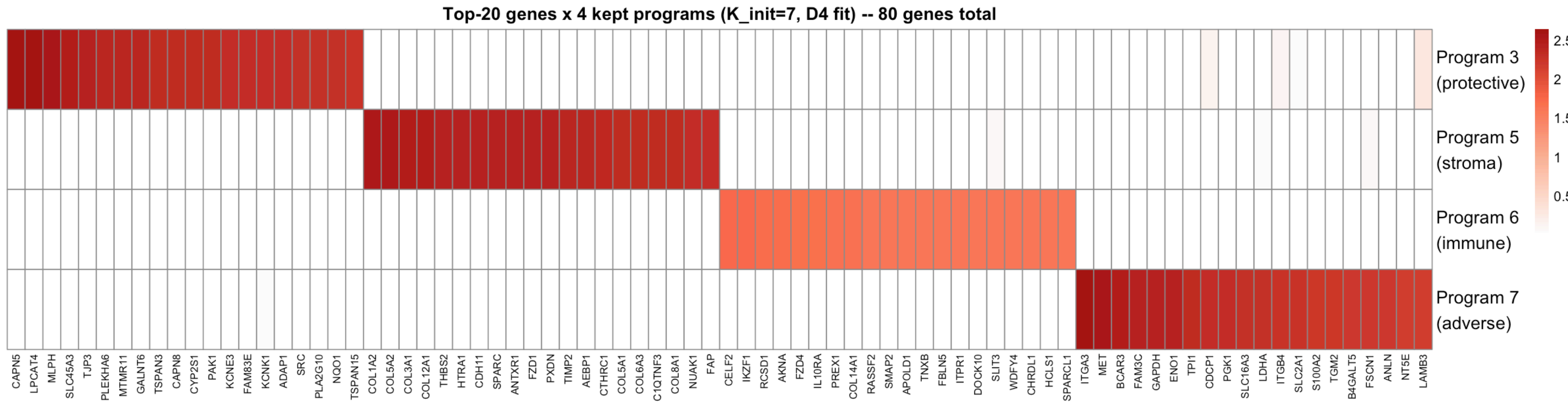

Figure 4: **Figure.** Top-20 genes by loading weight for each of $K_{\text{init}} = 7$'s 4 kept factors (the other 3, fully pruned, are omitted), rows = programs, columns = genes. Weights are **non-negative** (point-exponential prior); the block-diagonal structure means each program's top genes carry near-zero weight in the other 3.

- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus; see the sweep later) $\rightarrow$ **2 survival-active** + 2 genomics-only kept: Program 3 (**protective**) and Program 7 (**adverse**), classified by the sign of each program's fitted Cox coefficient $\hat{\beta}_k$ (positive $\rightarrow$ higher activation, shorter survival, adverse; negative $\rightarrow$ longer survival, protective)
- The 2 survival-active programs are characterized by **5 methods**: gene-set enrichment (fgsea, Fast Gene Set Enrichment Analysis, and ORA, Over-Representation Analysis), correlation with the PurIST classical/basal subtype score, hazard-ratio consistency across all 5 external cohorts, and gene-list overlap with the lab's published DeSurv factors
- The 2 genomics-only programs carry **weaker evidence**: their top-weighted genes were characterized by gene-set enrichment (fgsea) and overlap with the lab's published DeSurv factors, the only 2 of the 5 methods that apply without a survival signal — **Program 5** reads as **desmoplastic stroma** (7 collagen genes, SPARC, THBS2, FAP) and **Program 6** as **immune infiltrate** (IKZF1, AKNA, IL10RA, HCLS1); unlike Programs 3/7, their labels have not been tested for reproducibility across cohorts

On real data, the recommended fit starts at K_init=7 and ARD keeps 4 programs: 2 survival-active (Program 3 protective, Program 7 adverse) and 2 genomics-only (Program 5 stroma, Program 6 immune infiltrate). The figure shows only the 4 kept factors, not all 7; the 3 fully-pruned (dead) factors are omitted since they carry no real gene-program structure. Weights are non-negative because L and F use a point-exponential prior. On the biological labels: Programs 3 and 7 rest on five methods, including out-of-sample hazard-ratio replication across all five external cohorts, which is strong evidence. Programs 5 and 6 rest on two of those five methods agreeing with each other (gene-set enrichment and overlap with DeSurv's published factors), which is suggestive but meaningfully weaker; a formal external-cohort robustness check for THOSE two programs was run as a negative control (confirming they carry no survival signal, which they shouldn't) rather than as positive evidence for the stroma/immune labels themselves. If asked how confident we are in "desmoplastic stroma" and "immune infiltrate" specifically: say plainly that this is a preliminary read, not a validated claim, and flag testing whether the same two programs reappear on other cohorts as a future action item.

PDAC Cohorts: Prognostic Association

The two survival-active programs stratify patients (split at median **projection score** $(YF)_k$, the quantity the model scores on):

- **Adverse, Program 7** (higher activation $\rightarrow$ shorter survival), log-rank $p = 0.00047$
  - top genes: *ITGA3, MET, BCAR3, FAM3C, GAPDH, ENO1, TPI1, PGK1*
- **Protective, Program 3** (higher activation $\rightarrow$ longer survival), log-rank $p = 0.0077$
  - top genes: *CAPN5, LPCAT4, MLPH, SLC45A3, TJP3, PLEKHA6, MTMR11, GALNT6*

Proportional-hazards assumption not violated (Schoenfeld test).

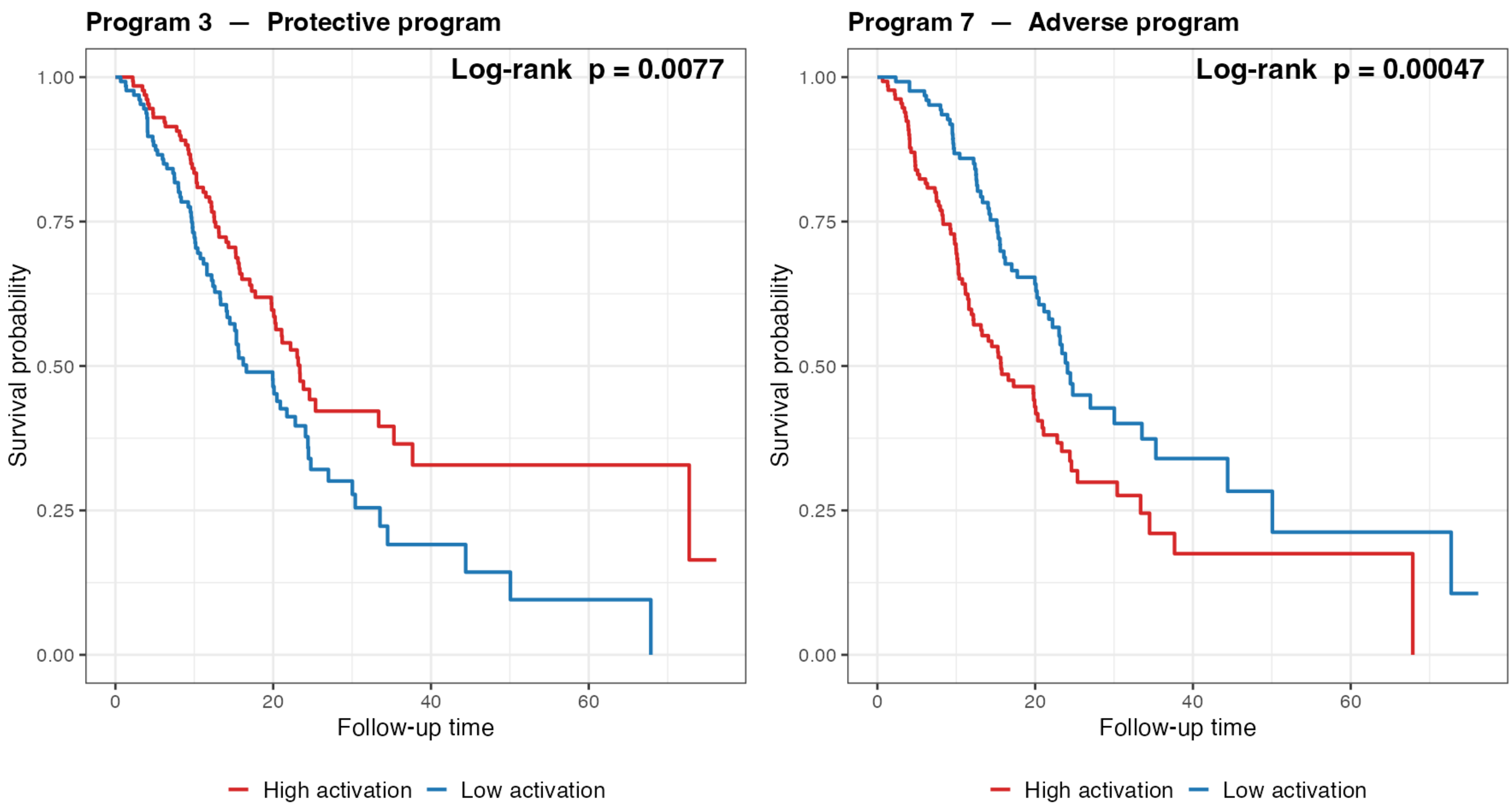

Figure 5: **Figure.** Kaplan–Meier curves for the two survival-active programs, patients split at the median projection score $(YF)_k$ in the training cohorts. The adverse program's high-activation group has worse survival; the protective program's, better. Direction read empirically from the curves.

The two active programs stratify patients into clear survival groups, split by the projection score (YF)_k. Program 7, the adverse program, carries MET, ITGA3, and glycolytic genes, biologically a classic aggressive signature. Program 3, protective, carries epithelial differentiation markers. Directions are read off the curves. Schoenfeld test shows PH holds.

Choosing K_{init}: the consensus sweep

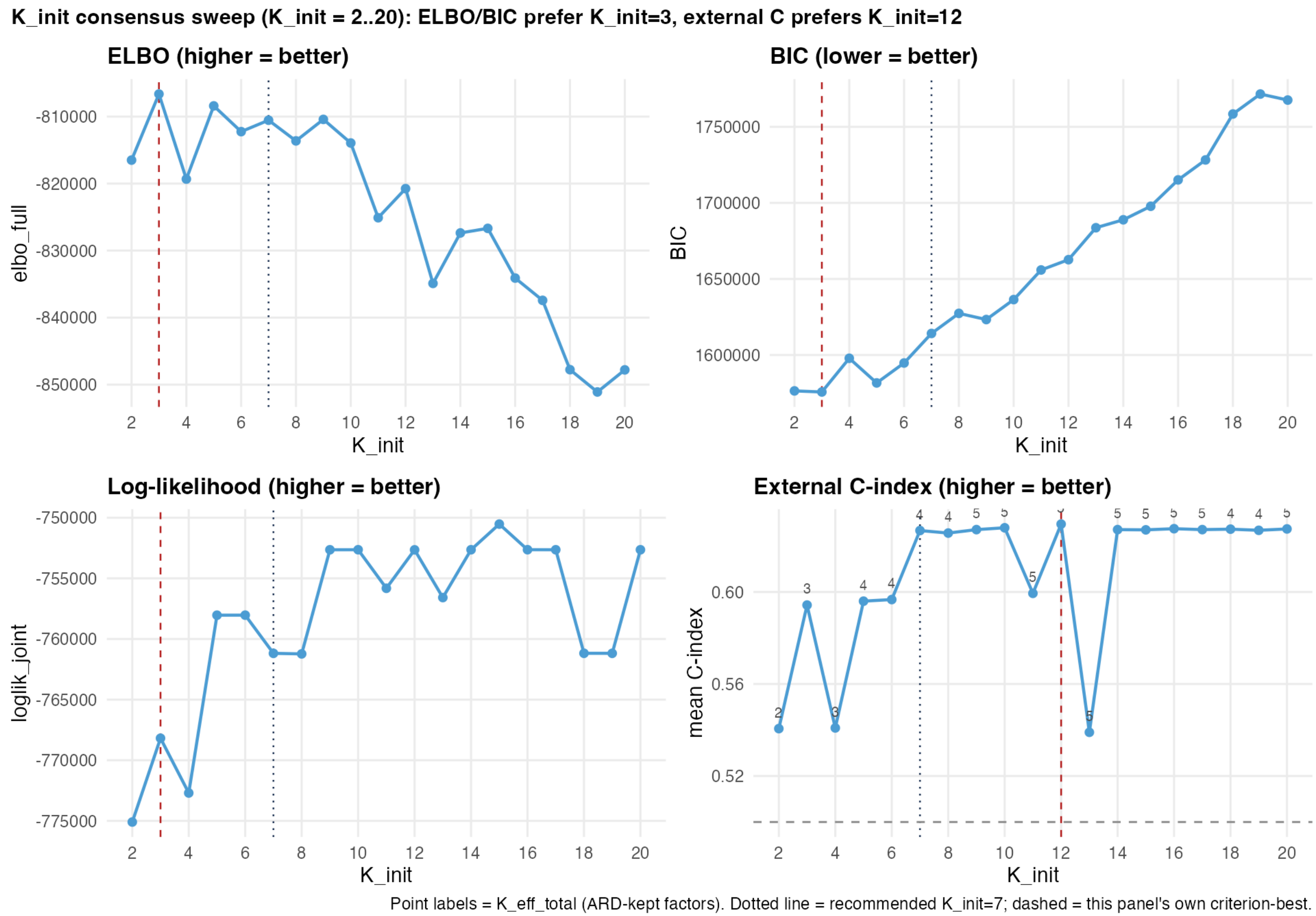

Figure 6: **Figure.** $K_{\text{init}} = 2\text{-}20$, each fit classified by ARD. Each panel states its own preferred direction; higher is better for ELBO, log-likelihood, and external C-index, lower is better for BIC. Point labels on the C-index panel are K_{eff}total (ARD-kept factors). Dotted line = recommended $K_{\text{init}} = 7$; dashed = each panel's own criterion-best.

- **ELBO- and BIC-preferred:** $K_{\text{init}} = 3$ (both favor the smallest K_{init} tested). **External-C-preferred:** $K_{\text{init}} = 12$ (favors more capacity). These do **not** agree, and that disagreement is shown here rather than retuned away.
- **Recommendation:** $K_{\text{init}} = 7$**.** Survival-active count is stable at 2 across a wide range of K_{init}; $K_{\text{init}} = 7$ is externally competitive (on the C-index plateau)

and the simplest single-fit path. Full table in the appendix.

Here is where the disagreement matters. ELBO and BIC agree with each other, preferring K_init=3, both favor small K_init: BIC's df penalty, tens of thousands of units per K_init here, dominates once K_init climbs into double digits, and ELBO simply peaks early on this training objective. External C-index prefers K_init=12, but K_init=7 through 20 mostly sit on a plateau around C~0.627, with two isolated dips at K_init=11 and 13. I want to show the actual curves and provoke the question of whether a different K_init in that plateau might be preferable, rather than presenting K_init=7 as the only defensible choice. K_init=7 is kept as the practical recommendation because the survival-active count is stable there and it's the simplest single-fit path.

Joint model vs. the unsupervised two-step

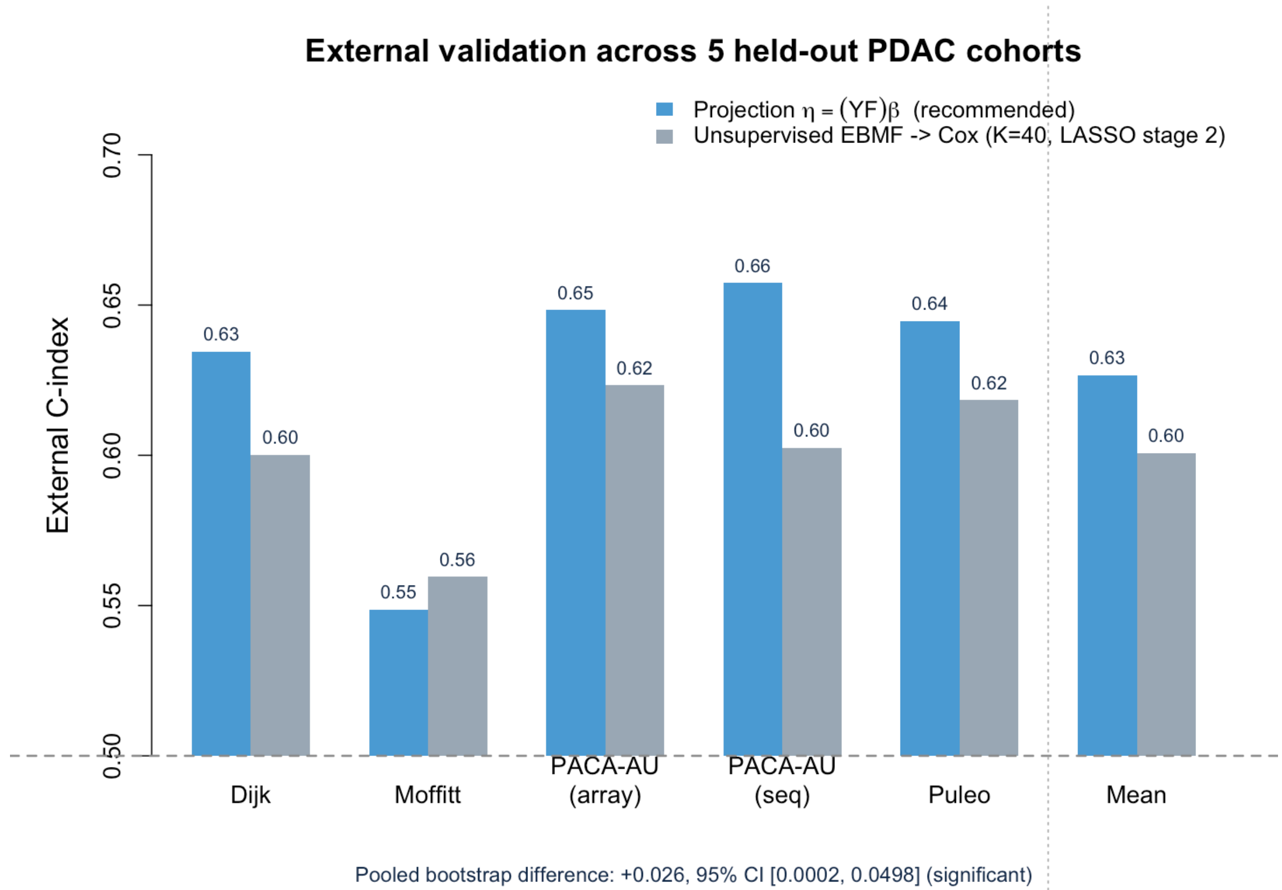

Figure 7: **Figure.** External C-index across the five held-out PDAC cohorts (+ mean): the recommended projection model vs. the fairest independent two-step baseline (EBMF → Cox, $K = 40$, LASSO stage 2). $K = 40$ is a large factor count the two-step method was never told YFB's answer to, with a regularized second stage rather than a small K or an unregularized `coxph()` that overfits at $K = 40$. Dashed line = chance.

- **Projection model mean C = 0.627** · unsupervised EBMF → Cox (K=40, LASSO) 0.601, across the five held-out cohorts

- Pooled bootstrap difference: **+0.026, 95% CI [0.0002, 0.0498]**, n=616 pooled across cohorts, **significant**
- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus) → $K_{\text{eff}} = 2$ survival-active of 4 kept

The real-data analog of the synthetic slide. Two arms: the recommended projection model, and the fairest independent two-step baseline I could build: EBMF at a large, YFB-uninformed K=40, with a regularized (LASSO) stage-2 Cox rather than a plain coxph() that overfits at that K. The pooled bootstrap difference is significant. This is not the only two-step comparison in the project; other configurations trail YFB by more, some without reaching significance at this sample size. This is the one built specifically to be hardest to argue with.

Model Comparison Summary

Model	Linear predictor	Preprocessing	K	K_eff	Real ext. C
Projection (recommended)	**(YF)**	**Per-platform z-std, survival-ranked genes**	**K_init=7**	**2**	**0.627**
Unsupervised two-step	EBMF → Cox	Per-platform z-std, survival-ranked genes	40	15/40 (LASSO)	0.601

Real ext. C = mean across 5 held-out PDAC cohorts. **K_init** = starting factor count, selected by an ELBO / BIC / log-likelihood / external-C consensus (Stage 1); **K_eff** = survival-active programs by ARD, $|\hat{\beta}| > 10^{-3}$ (Stage 2), for the projection model. For the two-step baseline, **K_eff** is the number of the 40 EBMF factors that keep a nonzero coefficient after LASSO-regularized, cross-validated stage-2 Cox fitting: a different mechanism from ARD (L1 shrinkage vs. an ARD prior on β), but it answers the same question of how many factors the model actually uses. Gene selection ranks genes by combined mean × variance per cohort.

- **Recommendation:** projection predictor $(YF)\beta$ · per-platform z-standardization · survival-ranked gene selection · **no** cohort indicator
- $K_{\text{init}} = 7 \to K_{\text{eff}} = 2$ survival-active (4 kept total), with **exact** external scoring

> **i** On gene selection
>
> The combined-rank selection recipe was **motivated by** the lab's penalized survival-MF (DeSurv) preprocessing; here it is applied within the fully Bayesian model.

Two rows: the recommended projection model against the unsupervised two-step baseline. Projection predictor with per-platform z-std and survival-ranked gene selection, K_init=7 selected by consensus, is the recommendation.

Impact / key findings

1. **Joint supervision beats the unsupervised two-step**, and costs nothing when there is no survival signal to exploit (signal-strength sweep).
2. **Preprocessing is decisive.** Per-platform z-standardization before merging is the difference between a working model and $\beta \to 0$; **10 of 12** alternatives collapse.
3. **K selection is a two-stage split.** Cross-validated external C-index, with ELBO, BIC, and log-likelihood, helps pick K_{init}; ARD shrinkage then determines the survival-active subset ($K_{\text{eff}} = 2$) directly from that single over-specified fit, with no separate per-K validation loop needed for K_{eff} itself.
4. **Over-specifying K_{init} and trusting ARD costs nothing.** Confirmed in simulation (15 seeds): recovery, specificity, and C-index are flat whether the model starts at the true K or 3x over-specified.
5. **Mean external C-index = 0.627** across five independent PDAC cohorts, fully Bayesian throughout (adaptive priors, no α tuning, uncertainty-aware).

Limitations

- **It is not obvious which K_{init} is best.** BIC and ELBO favor a much smaller K_{init} than external C-index does; that disagreement is shown on the sweep slide rather than resolved by picking whichever criterion agrees with us
- **The ARD retention thresholds (PVE $> 1\%$, $|\hat{\beta}| > 0.001$) are not literature-derived.** The $\hat{\beta}$ cutoff was reverse-engineered from this model's own coefficient scale, not taken from a principled sparse-factor-model source; a literature check and a sensitivity sweep are still open. A fixed threshold on $\hat{\beta}_k$ also assumes every factor's projection score is on a comparable scale: each factor's loading vector is unit-normalized (removing the dominant scale gap), but that does not guarantee equal projection-score *variance* across factors, which is a further open question
- **Small external cohorts give noisy estimates.** Per-cohort C ranges down to 0.55 (Moffitt), near chance
- The 2 survival-active programs are **statistically prognostic and biologically annotated by 5 independent methods** (Program 7 basal-like/adverse, Program 3 classical/protective); the 2 genomics-only programs' labels rest on weaker evidence (2 of those methods, not externally validated) and a matched head-to-head against DeSurv on one protocol both remain open

Five takeaways, then a limitations section that names the same open questions raised on the slides that produced them, not new ones: the K-selection criteria disagree with each other, and it isn't obvious which K_init is actually best; small external cohorts give noisy per-cohort estimates; the ARD thresholds aren't literature-grounded; and the biological annotation is solid for the two survival-active programs but preliminary for the two genomics-only ones. On the beta-comparability point specifically, if asked: each factor's loading vector F_k is unit-normalized before the projection score is computed, both at train and test time, which removes the dominant scale gap (raw F_k norms grow like sqrt(p), so without this an unnormalized factor loading on more genes would mechanically get a larger projection score regardless of survival relevance). What it does NOT do is guarantee every factor's projection score has the same variance across patients, since that still depends on which genes each factor loads on and how correlated they are. So the fixed |beta_hat|>0.001 threshold is applied on a partially, not fully, equalized scale. This was implemented for fitting stability, not explicitly to solve cross-factor comparability, so nobody has checked whether it's sufficient. Flagged as an open item, not swept under the rug.

Future directions & summary

Current status

- K **selection**: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_{init}; ARD shrinkage determines K_{eff}
- **Gene programs biologically annotated**: the 2 survival-active programs by the 5 methods on the gene-programs slide; the 2 genomics-only programs more tentatively, by 2 of those methods

Next steps

1. **Multiple cohorts / multiple modalities via indicator columns in** L: stack methylation and expression, possibly with modality-specific priors; leave-one-study-out validation; train on A vs. B vs. A+B
2. **Manuscript preparation**: methods-primary framing, PDAC as the application

Summary

1. Joint supervised factorization via the projection predictor $(YF)\beta$; **point-exponential priors on** L, F**, a normal prior on** β, non-parametric baseline hazard
2. **Preprocessing is crucial**: per-platform normalization prevents the survival precision from being dominated and β from collapsing
3. K **selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform** $K_{\textbf{init}}$**; ARD shrinkage determines** $K_{\textbf{eff}}$, validated on real data (five external PDAC cohorts, mean C = 0.627) and synthetic data with known ground truth
4. **Recommended model**
   - Linear predictor: projection $(YF)\beta$
   - L cohort indicator: No (candidate direction for multi-cohort/multi-modality extension, above)
   - Pre-processing: per-platform z-std
   - $K_{\text{init}} = 7 \rightarrow K_{\text{eff}} = 2$, mean external C = **0.627**

K selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_init; ARD shrinkage determines K_eff. The two survival-active programs are solidly annotated; the two genomics-only ones more tentatively so, worth being upfront about if asked. The forward-looking direction I want to lead with is extending L with indicator columns for multiple cohorts and/or multiple modalities: stack methylation and expression, possibly with modality-specific priors, and validate with leave-one-study-out and train-on-A-vs-B-vs-A+B comparisons. Then manuscript prep.

Discussion

Thank You

Questions, suggestions, and feedback welcome!

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Open the floor. Appendix slides cover convergence diagnostics, survival-activity magnitudes, and the DeSurv comparison.

Appendix

Appendix: K_{init} sweep, full table

K_init	Surv-active	Genomics-only	Dead	K_eff (kept)	ELBO	BIC	Log-lik	External C
2	1	1	0	2	-816,495	1,576,416	-775,093	0.5406
3	1	2	0	3	-806,636	1,575,712	-768,183	0.5943
4	1	2	1	3	-819,328	1,597,855	-772,698	0.5409
5	3	1	1	4	-808,404	1,581,651	-758,038	0.5960
6	3	1	2	4	-812,248	1,594,760	-758,035	0.5967
7	**2**	**2**	**3**	**4**	**-810,540**	**1,614,174**	**-761,185**	**0.6267**
8	2	2	4	4	-813,629	1,627,379	-761,230	0.6256
9	2	3	4	5	-810,421	1,623,321	-752,643	0.6271
10	2	3	5	5	-813,944	1,636,429	-752,640	0.6279
11	3	2	6	5	-825,096	1,655,903	-755,819	0.5994
12	2	3	7	5	-820,752	1,662,678	-752,649	0.6295
13	3	2	8	5	-834,897	1,683,664	-756,585	0.5390
14	2	3	9	5	-827,366	1,688,890	-752,641	0.6271
15	2	3	10	5	-826,682	1,697,778	-750,527	0.6270
16	2	3	11	5	-834,100	1,715,115	-752,638	0.6275
17	2	3	12	5	-837,425	1,728,238	-752,642	0.6271
18	2	2	14	4	-847,771	1,758,428	-761,180	0.6273
19	2	2	15	4	-851,099	1,771,549	-761,183	0.6268
20	2	3	15	5	-847,800	1,767,584	-752,643	0.6274

Highlighted row = the recommended $K_{\text{init}} = 7$. External C = mean C-index across the 5 held-out PDAC cohorts. BIC and Log-lik are ELBO-style bounds, not exact marginal-likelihood quantities.

The full sweep table, for anyone who wants the actual numbers rather than just the curves. Note the K_eff (kept) column is not monotone in K_init – it moves between 2 and 5 as ARD prunes differently at each starting point, and two rows (K_init=11, 13) show a visible external-C dip against their neighbors.

Appendix: simulation detail, factor recovery & specificity

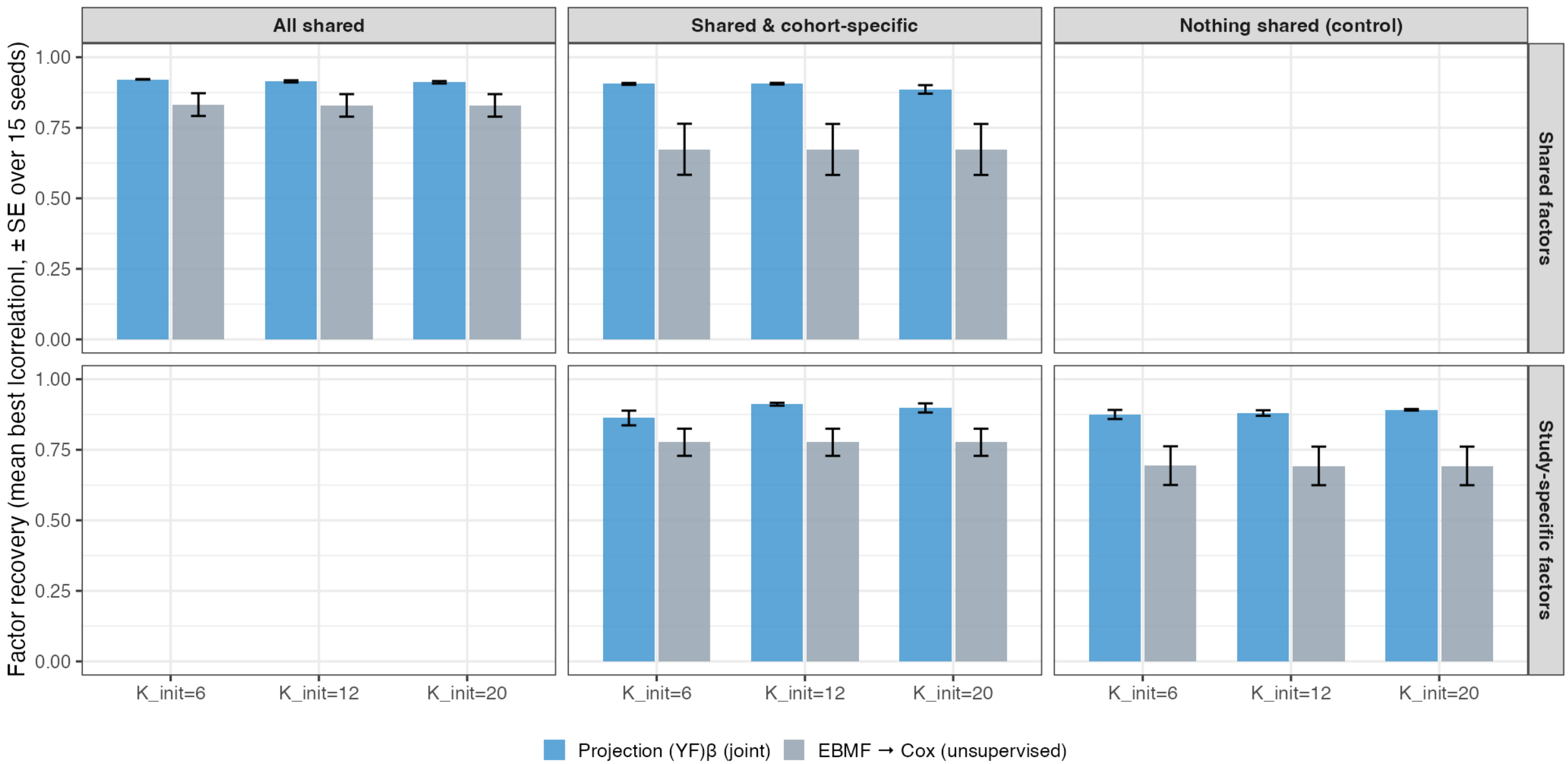

Figure 8: **Figure A.** Factor recovery (mean best |correlation| vs. the true gene programs) by K_{init}, arm, and scenario (mean ± SE, 15 seeds).

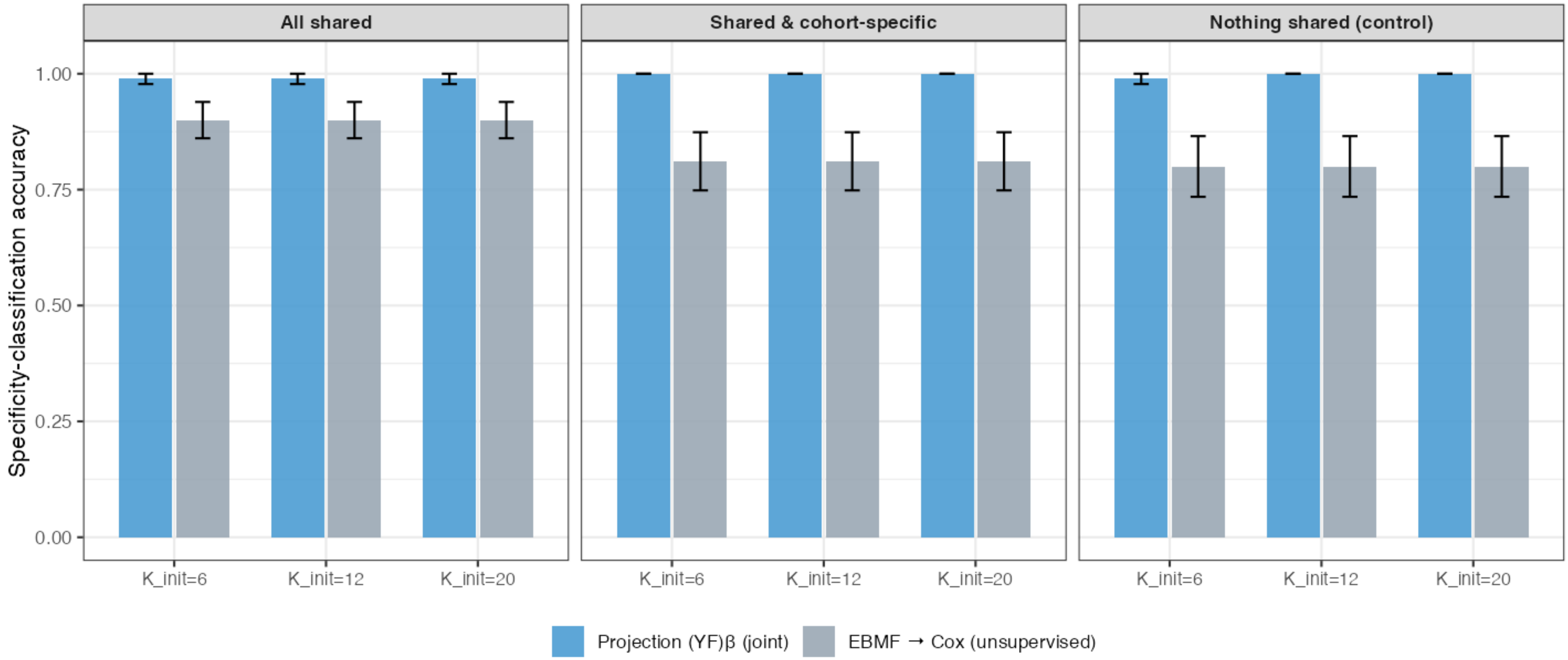

Figure 9: **Figure B.** Specificity-classification accuracy (fraction of factors correctly typed shared vs. study-specific) by K_{init} and scenario (mean ± SE, 15 seeds).

- The same flat-across-K_{init} pattern shown by C-index in the main results holds for recovery and specificity too, in every scenario

Backup detail for the "over-specifying K_init doesn't hurt" slide, in case someone wants to see recovery and specificity, not just C-index. Same flat story across K_init=6/12/20 in every scenario.

Appendix: convergence & survival-activity magnitudes

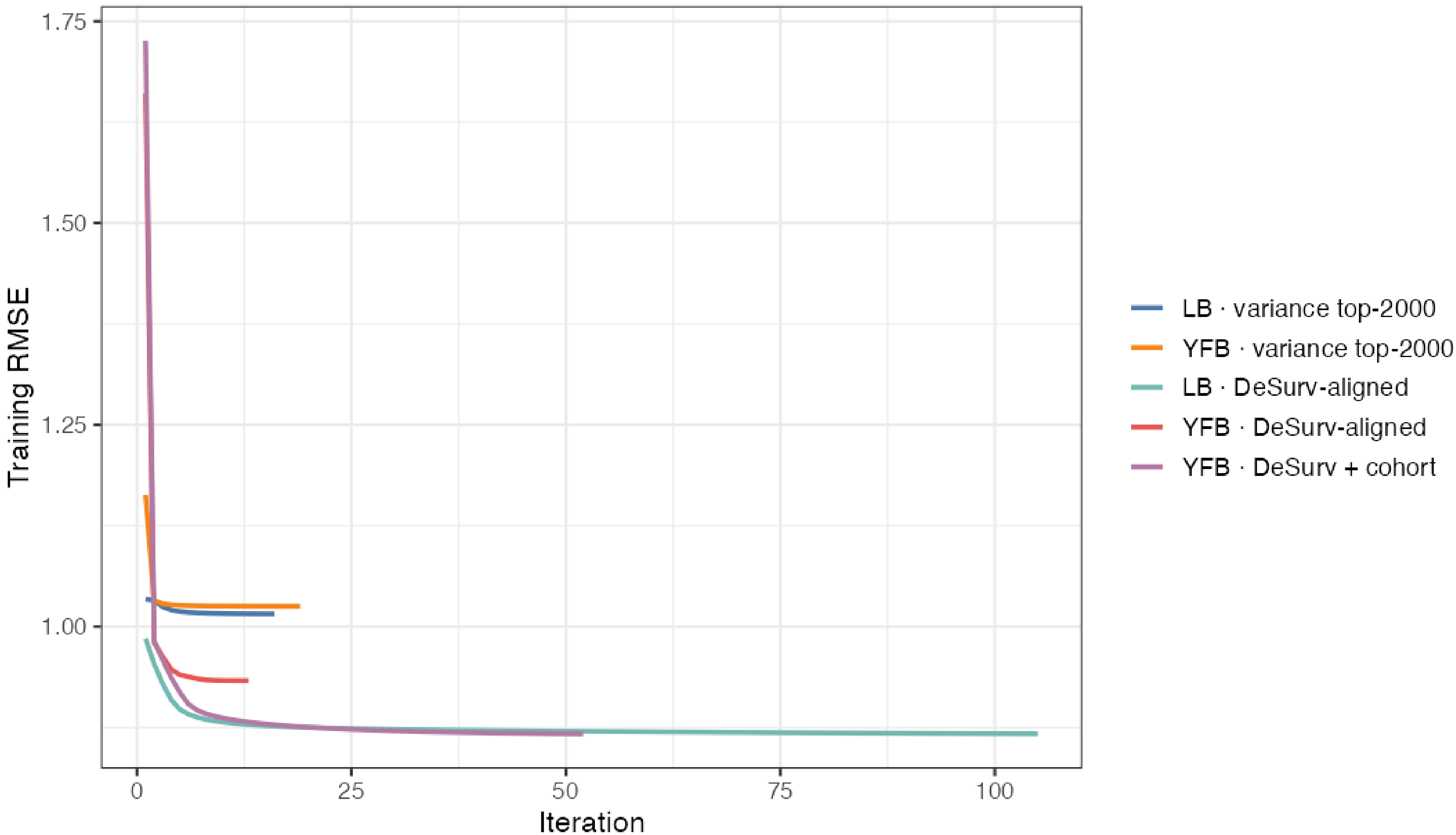

Figure 10: **Figure A1.** Training RMSE convergence traces across model configurations. RMSE stabilizes to a plateau; stopping is governed by the relative ELBO change falling below 10^{-5} ($\Delta L, \Delta\beta$ tracked as diagnostics).

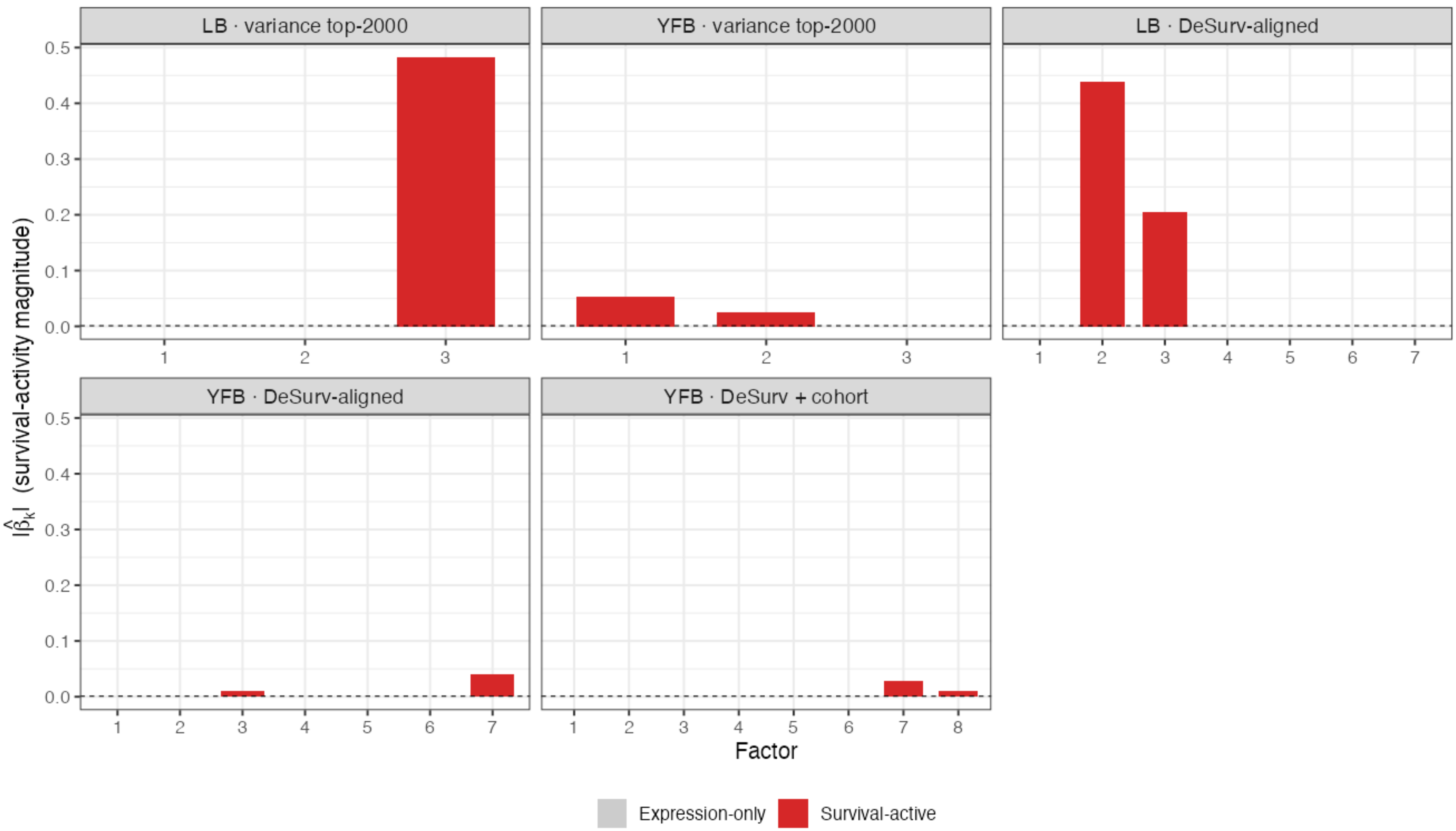

Figure 11: **Figure A2.** Survival-activity magnitude $|\hat{\beta}_k|$ per program. Most programs fall below the activity threshold (expression-only); a small number clear it (survival-active).

- The spike-and-slab / shrinkage behavior is visible: most programs are shrunk below the survival-activity threshold, with only the survival-active ones retained

Appendix: relation to DeSurv (Young et al., PNAS 2026)

Same design and same data: SBMF is the Bayesian counterpart. Both jointly fit supervised NMF + Cox with **non-negative gene programs**, share the same training set (**TCGA + CPTAC,** $n = 273$) and the **same five external PDAC cohorts** (Dijk, Moffitt, PACA-AU array/seq, Puleo; $n = 616$), score new patients by **projection** (no retraining), and use the **combined-rank top-3000-per-cohort** gene selection. Both also find that the **highest-variance program is *not* the most prognostic**.

DeSurv objective: $(1-\alpha)\,\mathcal{L}_{\text{NMF}} - \alpha\,\mathcal{L}_{\text{Cox}}$, with the survival gradient on the gene programs W.

What SBMF changes

	DeSurv	SBMF
Estimation	penalized optimization	Bayesian variational (CAVI)
K selection	Bayesian-optimization over k, α, λ jointly	ARD shrinkage + ELBO/BIC/log-likelihood/external-C consensus on K_{init} only
Shrinkage	fixed elastic-net	empirical-Bayes priors (learned)
Uncertainty	point estimates	posterior mean + variance
Noise	least-squares	per-gene precision τ; $E[L^2]$ correction

- **DeSurv:** $k = 3,\ \alpha = 0.334$; pooled external **HR/SD** $= 1.50$ (95% CI 1.31–1.72)
- **SBMF (YF) :** $K_{\text{init}} = 7 \to K_{\text{eff}} = 2$; mean external **C $=0.627**

Caveat: the two report different primary metrics (DeSurv: pooled HR/SD; SBMF: mean external C-index) and use slightly different normalization (DeSurv within-subject rank $\to$ 1970 genes; SBMF per-platform z-std $\to$ 2064). A **matched same-protocol head-to-head is a planned next step**; DeSurv is *not* re-run here.

This is the backup for the inevitable "how does this compare to DeSurv?" question. SBMF is the Bayesian sibling of DeSurv, same supervised NMF+Cox idea, same training data and the same five external cohorts, same projection-based scoring, and SBMF borrowed DeSurv's combined-rank gene selection. The genuine differences: DeSurv tunes k, alpha, and lambda jointly by Bayesian optimization; SBMF selects K_init by a consensus of fit/prediction metrics and lets ARD prune within that fit, with empirical-Bayes priors learning the shrinkage and no alpha or lambda to tune, plus posterior uncertainty and a heteroscedastic noise model. On numbers, DeSurv reports pooled HR per SD of 1.50; SBMF reports mean external C of 0.627, different metrics, so a matched head-to-head on one protocol is the right next step, not a claim of superiority today.

Appendix: the 5 gene-program validation methods, in detail

Method	Implementation	What it tests	Result
fgsea (Fast Gene Set Enrichment Analysis, primary)	Genes ranked by weight within a program; tested against Hallmark, Reactome, KEGG, GO-BP, and a custom PDAC collection (Moffitt/Bailey signatures + DeSurv's 3 published factors). BH-adjusted $p < 0.10$.	Do a program's top-weighted genes cluster into known biological pathways, more than chance?	3 dead programs: zero hits (consistency check). Programs 5/6: many hits — stroma/EMT (5), immune activation (6).
ORA (Over-Representation Analysis, sensitivity check)	Hypergeometric over-representation on the top N genes ($N \in \{50, 100, 150\}$) vs. the 2,064-gene background.	Does a simpler, threshold-based test agree with fgsea's ranked approach?	Confirms fgsea's pathway calls — not an artifact of the ranking method.

Method	Implementation	What it tests	Result
PurIST subtype concordance	Spearman correlation + Kruskal–Wallis of each program's patient loadings against the published PurIST classical/basal score, TCGA-PAAD ($n = 144$, external, out-of-sample).	Does the program track an independently validated, previously published PDAC subtype axis?	Programs 3/7 at strong opposite ends ($\rho = -0.575$ / $+0.582$) — recovers a known PDAC finding. Program 6 orthogonal ($\rho = -0.026$, $p = 0.75$); Program 5 weak basal lean ($\rho = 0.299$, $p = 2.8 \times 10^{-4}$).
DeSurv gene-list overlap	Jaccard index + hypergeometric p against each of DeSurv's 3 published factor gene lists, top 270 genes/program.	Do these programs replicate the lab's own prior published factors?	The 4 kept programs recover DeSurv's 3-factor structure; DeSurv's single combined stromal/immune factor splits into Programs 5 and 6.
External cohort robustness	Each program's leading-edge signature scored in all 5 held-out cohorts; univariate Cox fit per cohort.	Does the prognostic direction replicate out-of-sample?	Confirmatory for 3/7 (direction replicates across cohorts). Negative control for 5/6 (no survival signal, as expected — not evidence for their labels).

Programs 3 and 7 (survival-active) are supported by all 5 methods, including 2 genuinely out-of-sample checks (PurIST in TCGA-PAAD, hazard-ratio replication across 5 cohorts). Programs 5 and 6 (genomics-only) are supported by only fgsea + DeSurv overlap — plausible readings, not yet tested for cross-cohort reproducibility.

Backup detail for Slide 17's "5 methods" claim, in case someone wants the actual definitions and numbers rather than the one-line summary. fgsea and ORA are both gene-set enrichment tests (fgsea ranks all genes by weight; ORA just thresholds to a top-N list as a simpler robustness check on fgsea). PurIST is an independent, previously published PDAC subtype classifier — using it here is a genuinely out-of-sample check, not something derived from our own data. DeSurv overlap checks replication against the lab's own prior published factors. External cohort robustness is the other genuinely out-of-sample check: does the prognostic direction hold up in 5 cohorts the model never trained on. Emphasize the asymmetry: Programs 3/7 have both out-of-sample checks plus all the internal ones; Programs 5/6 only have the 2 internal checks, which is exactly why their labels are flagged as preliminary on Slide 17.